

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 5

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I M Vishnu Vardhan Reddy(192572062)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Vishnu

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded

ASSIGNMENT 3

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Cloud Computing Lab Record

Experiment 1: Mark Sheet Processing System (SaaS)

Aim

To create a simple cloud-based Mark Sheet Processing System using a Cloud Service Provider (Google Forms + Google Sheets) to demonstrate Software as a Service (SaaS).

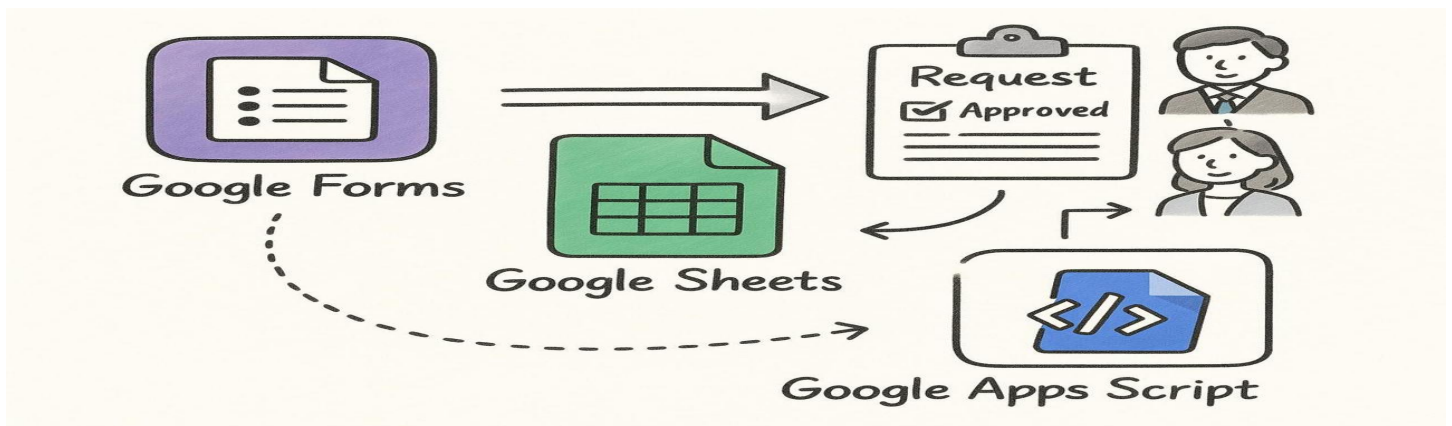
Cloud Service Provider

- Google Forms
- Google Sheets

Required Fields

- Student Name
- Register Number
- Department
- Subject 1
- Subject 2
- Subject 3
- Subject 4
- Subject 5
- Total Marks
- Average
- Percentage
- Rank
- Overall Performance

Architecture



Procedure

1. Open Google Forms.
2. Create a new blank form.
3. Title it Mark Sheet Processing System.
4. Add all required fields.
5. Link the form to Google Sheets.
6. Use formulas to calculate Total, Average and Percentage.
7. Sort the sheet to determine Rank.
8. Display Overall Performance.

Google Sheets Formula

Calculation Formula

Total =SUM(D2:H2)

Average =AVERAGE(D2:H2)

Percentage =I2/500*100

Rank =RANK(J2,\$J\$2:\$J\$20,0)

Performance Formula

=IF(J2>=90,"Excellent",
IF(J2>=75,"Very Good",
IF(J2>=60,"Good",
"Needs Improvement"))))

Sample Output

Name Reg No Average % Rank Performance

Vishnu 220123 92 92 1 Excellent

Rahul 220124 84 84 2 Very Good

Sample Interface

Mark-Sheet Calculator

Hindi		(Max : 100)
English		(Max : 100)
Maths		(Max : 100)
Science		(Max : 100)
Art		(Max : 100)
Computer		(Max : 100)
Total		
Percentage		
Grade		
	Calculate	

Screenshots to Upload

- Google Form
- Form Fields
- Linked Google Sheet
- Formula Calculations
- Final Result Table

Result

The Mark Sheet Processing System was successfully created using Google Forms and Google Sheets, demonstrating SaaS.

Experiment 2: Property Buying & Rental System (Chennai)

Aim

To create a cloud-based Property Buying & Rental System using Google Forms and Google Sheets.

Cloud Service Provider

- Google Forms
- Google Sheets

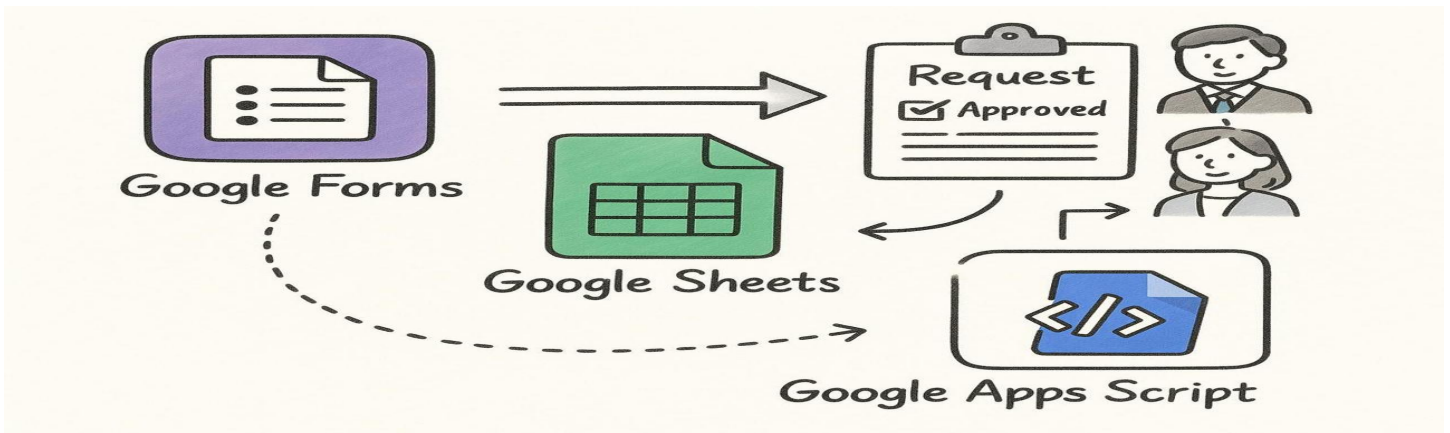
Required Fields

- Buy / Rent
- Residential / Commercial
- Owner Name
- Customer Name
- Mobile Number
- Area (Sq.ft)
- Place (Chennai)
- Budget Range
- Property Value
- Rental Agreement
- Duration
- Advance Amount

Chennai Locations

- Anna Nagar
- T Nagar
- Velachery
- OMR
- Adyar
- Tambaram
- Porur
- Sholinganallur

Architecture



Procedure

1. Open Google Forms.
2. Create Property Buying & Rental System.
3. Add dropdowns for Buy/Rent and Residential/Commercial.
4. Add Chennai location options.
5. Link to Google Sheets.
6. Store responses automatically.

Sample Form

Real Estate Enquiry Form

Please fill out the form below to know more about the property.

Your Name*

Address

Street Address

Street Address Line 2

City Region

Postal / Zip Code Country

Phone Number*

###

E-mail*

ex: myname@example.com

What kind of property are you interested in? *

Sample Data

Type	Area Place	Value
Buy	1200 Anna Nagar	₹85 Lakhs
Rent	900 Velachery	₹25,000
Commercial	1500 OMR	₹1.2 Crores

Features

- Online property booking
- Automatic data collection
- Cloud storage
- Easy filtering
- Multi-user access

Screenshots to Upload

- Property Form
- Chennai Location Dropdown
- Google Sheets Responses
- Stored Property Records

Result

The Property Buying & Rental System successfully demonstrated SaaS using cloud-based Google services.

Experiment 3: Type-2 Hypervisor Installation and VM Creation

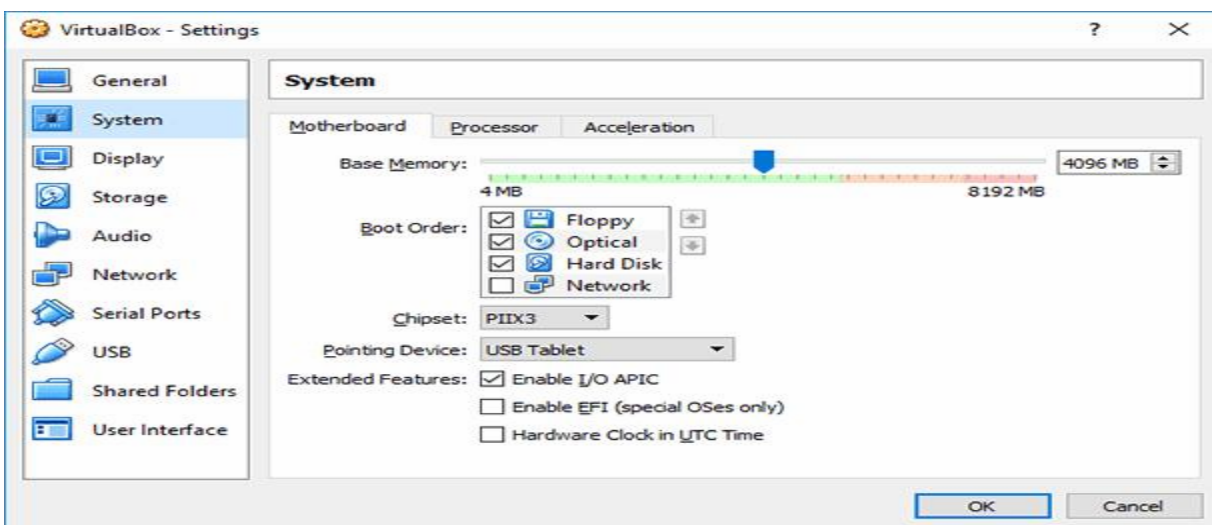
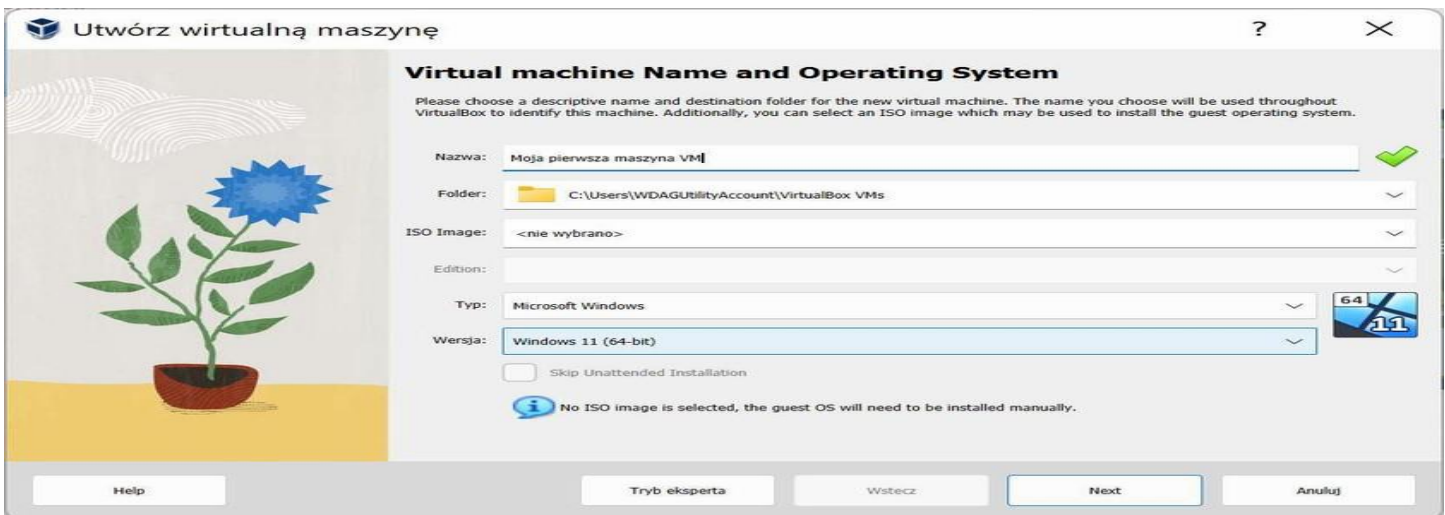
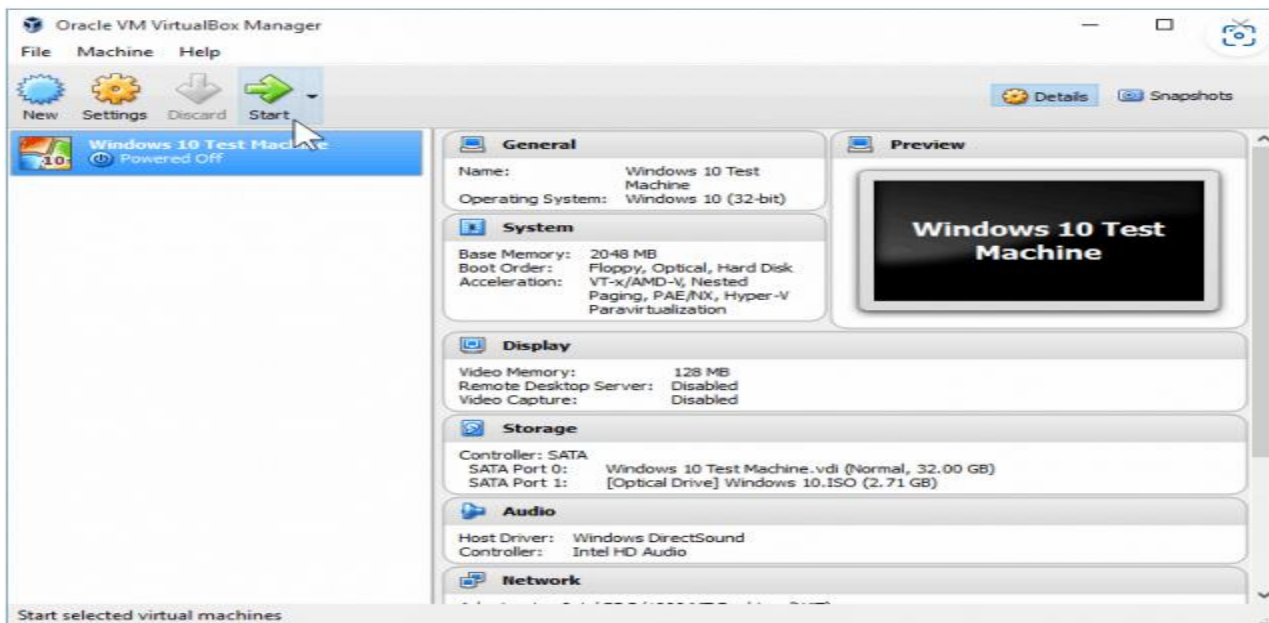
Aim

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a Virtual Machine.

Hypervisor

Oracle VM VirtualBox

(You may also use VMware Workstation.)



5

Host System

- Windows 11

Guest OS

- Ubuntu Linux

Required Configuration

Setting Value

CPU 5

RAM 12 GB

Disk 8 GB

Procedure

Step 1

Download and install VirtualBox.

Step 2

Open VirtualBox.

Step 3

Click New.

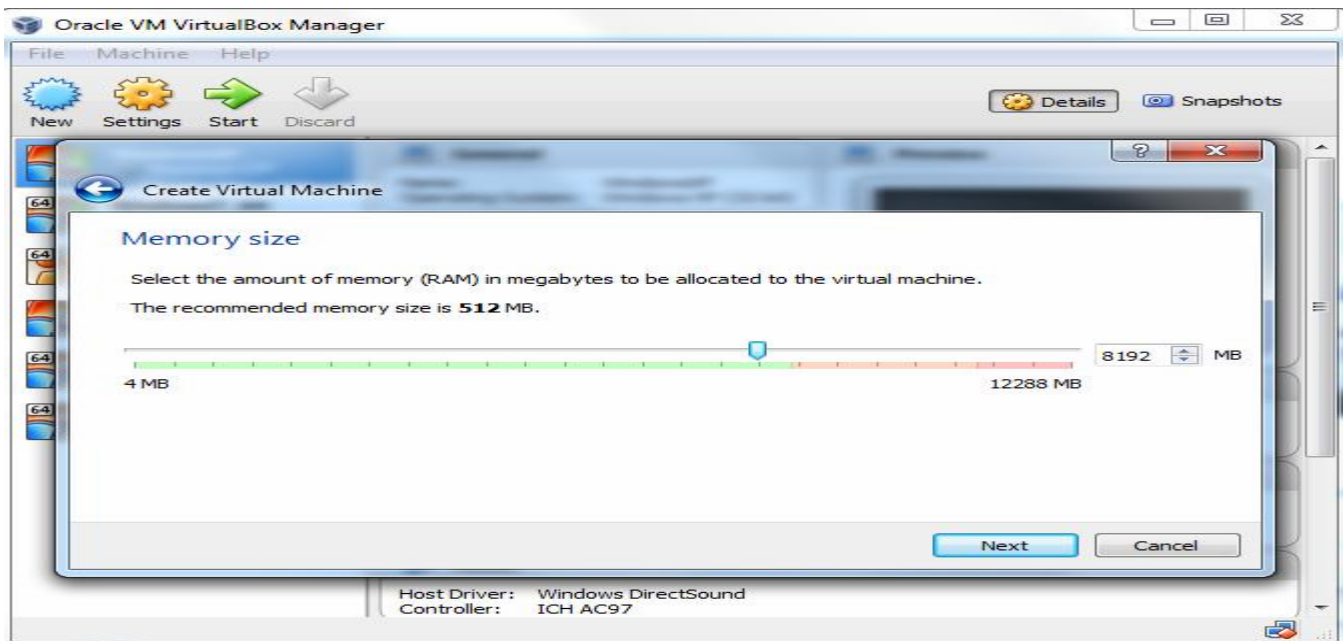
Step 4

Enter:

- Name: Ubuntu VM
- Type: Linux
- Version: Ubuntu (64-bit)

Step 5

Allocate Memory.

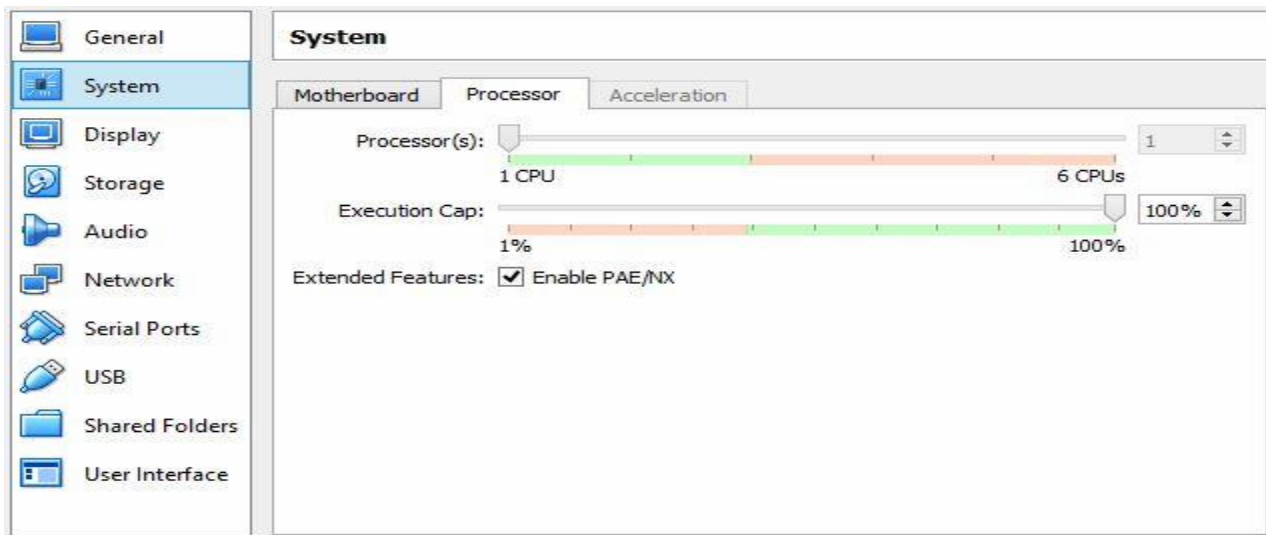


Set:

- 12288 MB

Step 6

Processor Allocation

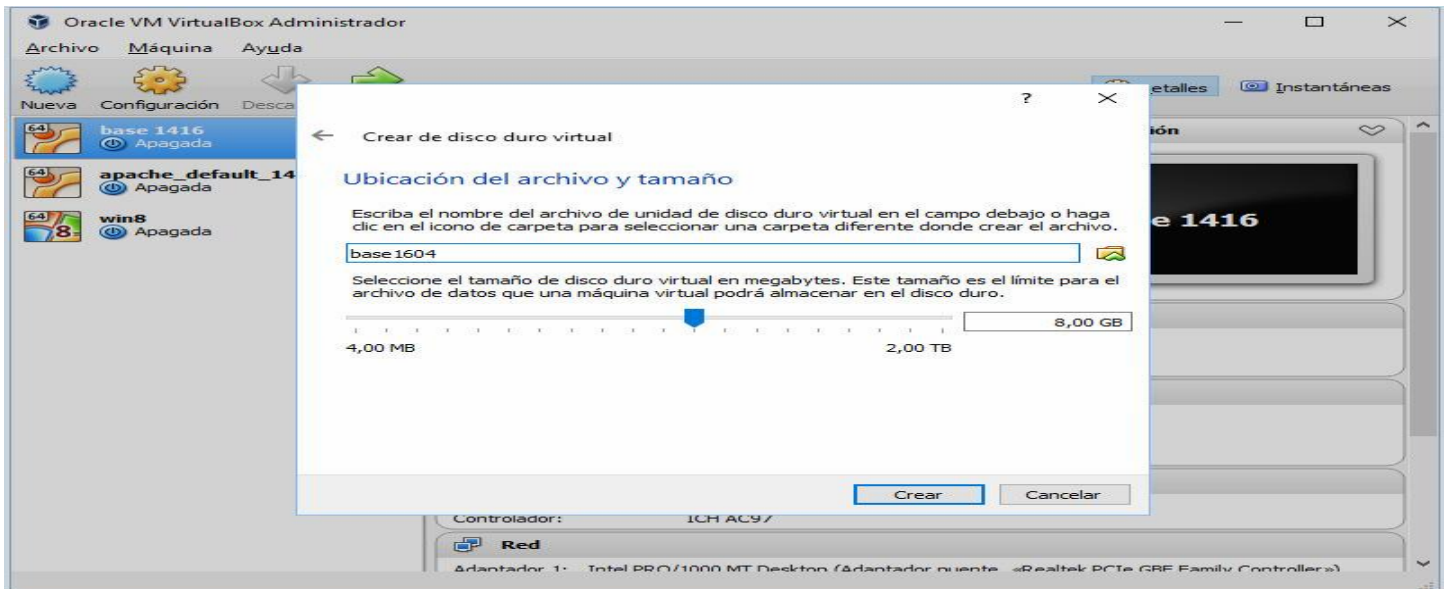


Set:

- 5 CPUs

Step 7

Create Virtual Hard Disk

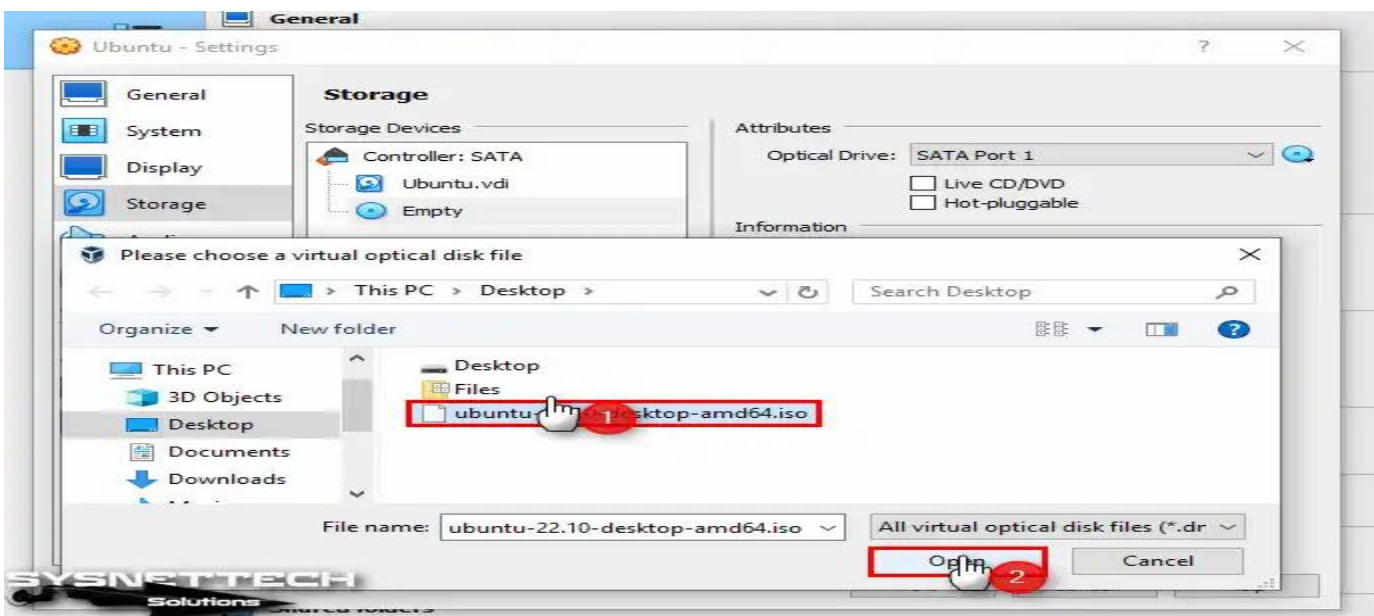


Set:

- VDI
- Dynamically Allocated
- 8 GB

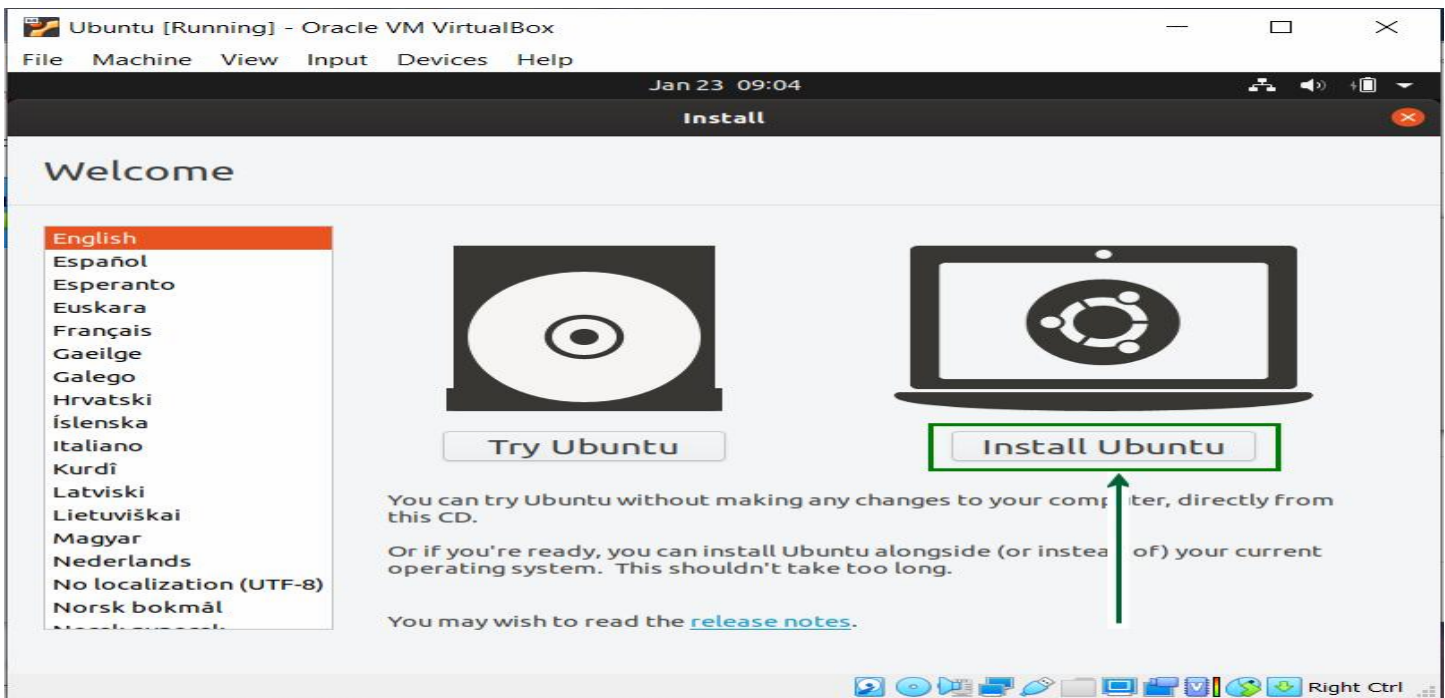
Step 8

Attach Ubuntu ISO.



Step 9

Start VM.



Screenshots to Upload

- VirtualBox Home
- New VM Wizard
- Memory Allocation
- CPU Allocation
- Storage Allocation
- Ubuntu Installation
- Running VM

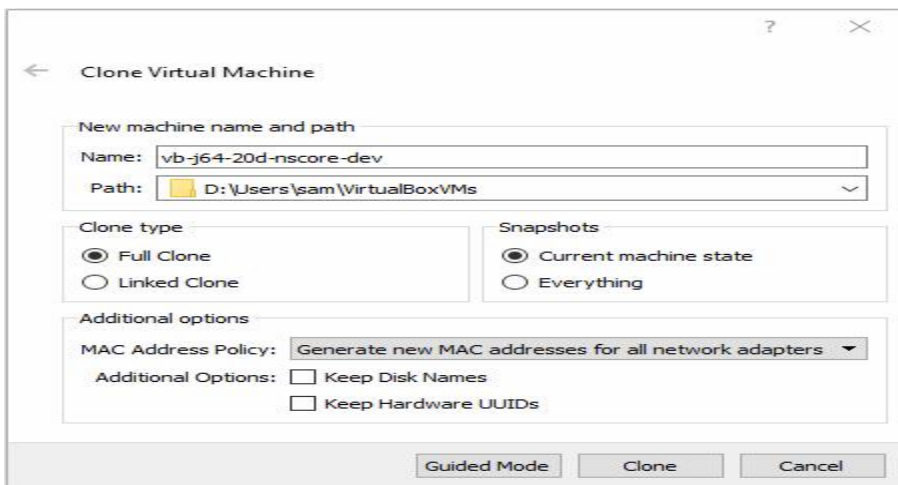
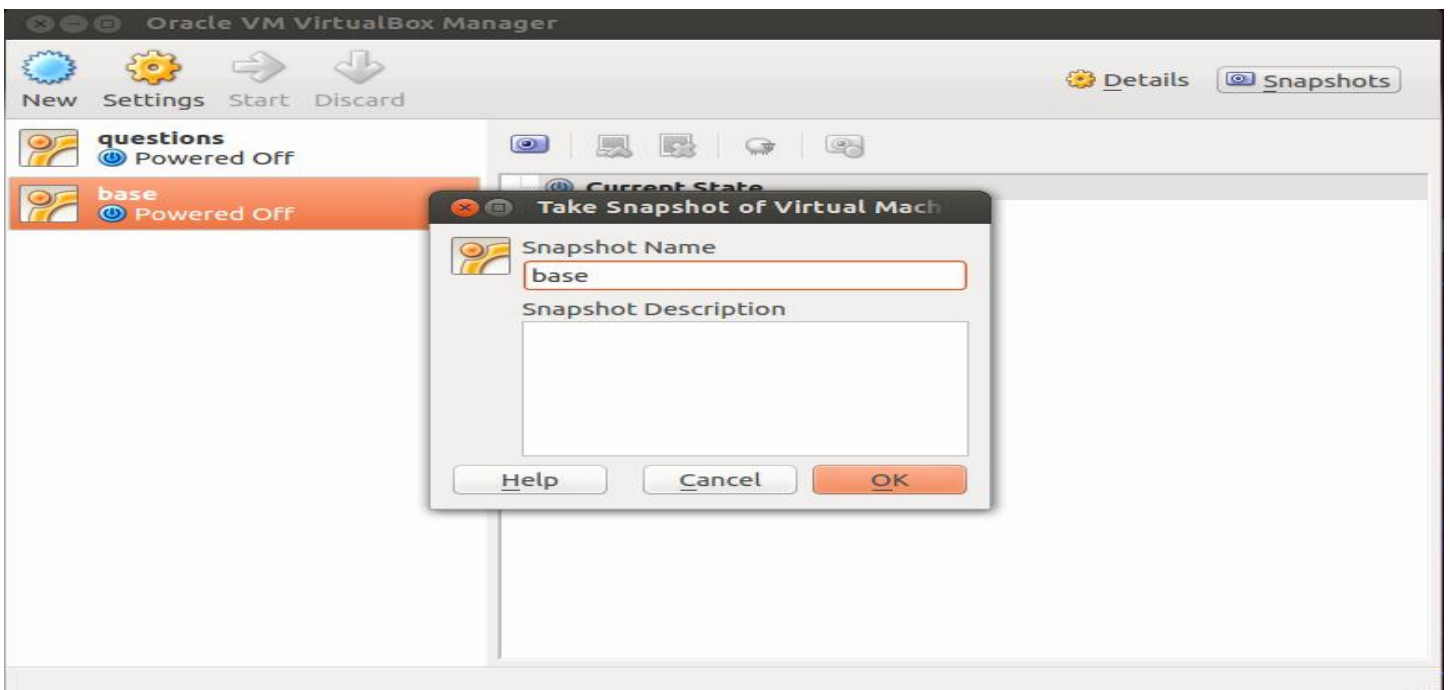
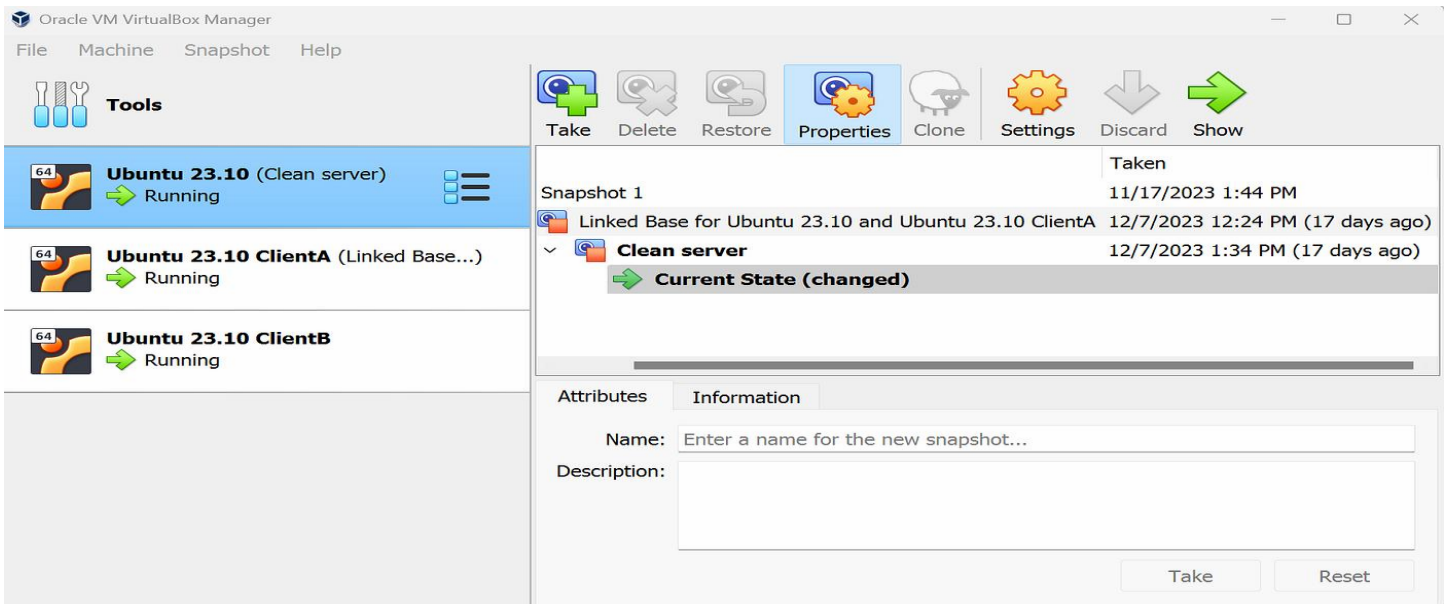
Result

A Virtual Machine with 5 CPUs, 12 GB RAM and 8 GB Storage was successfully created.

Experiment 4: Snapshot and Clone Demonstration

Aim

To create and restore a Virtual Machine Snapshot and create a cloned VM.



Procedure

Step 1

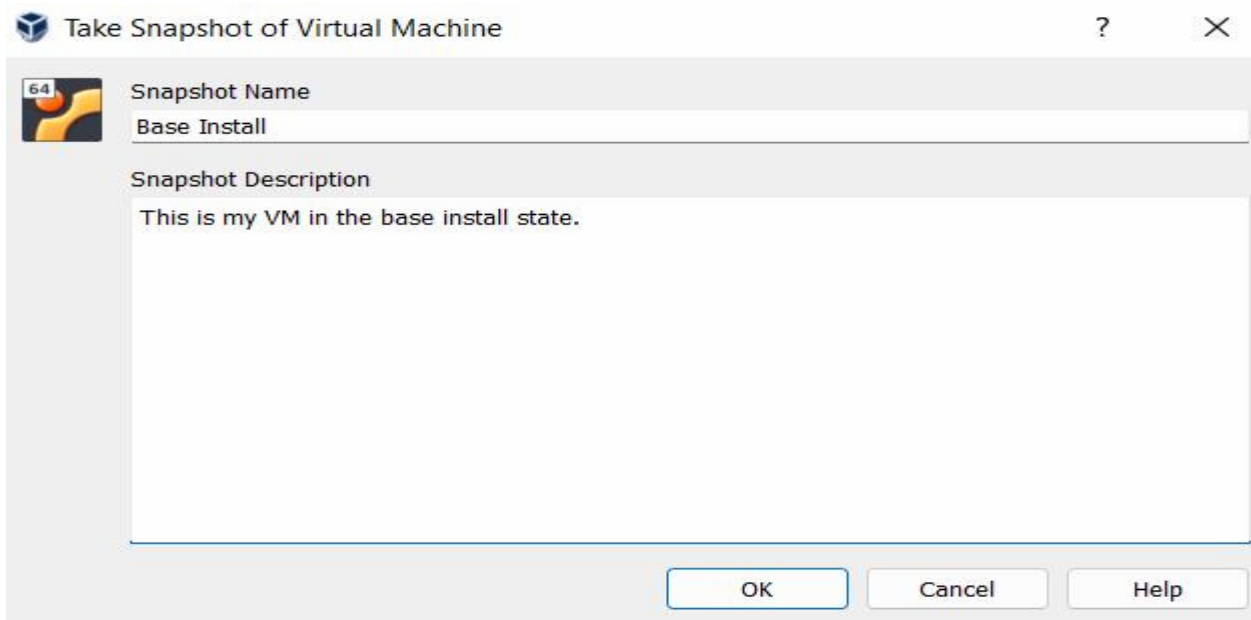
Open VirtualBox.

Step 2

Select Ubuntu VM.

Step 3

Take Snapshot.



Enter:

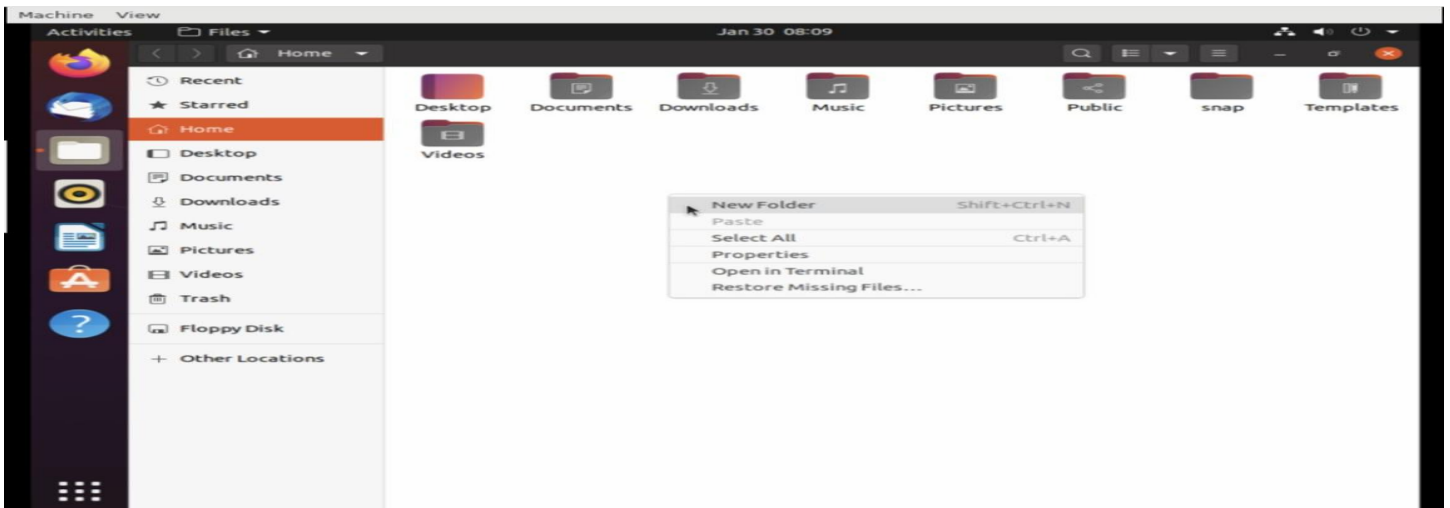
- Snapshot Name: Before Update

Step 4

Make Changes

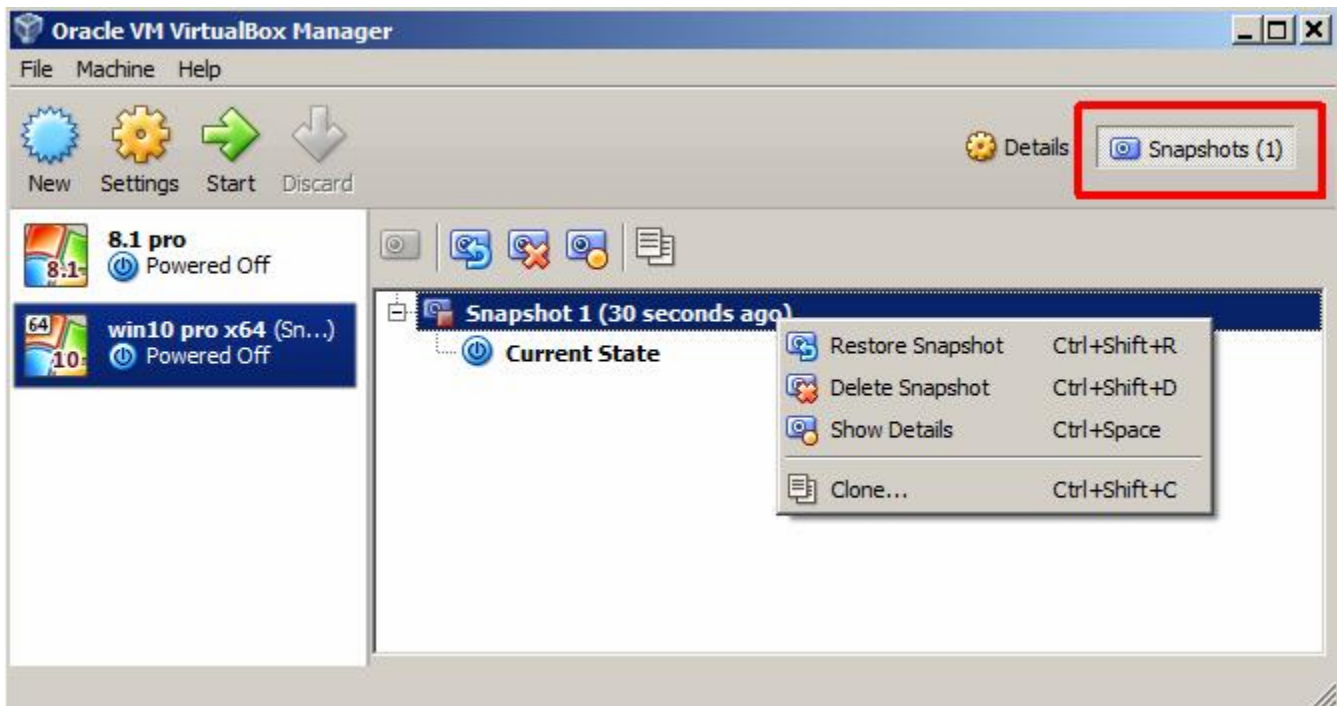
For example:

- Create a folder.
- Install a small application.



Step 5

Restore Snapshot.

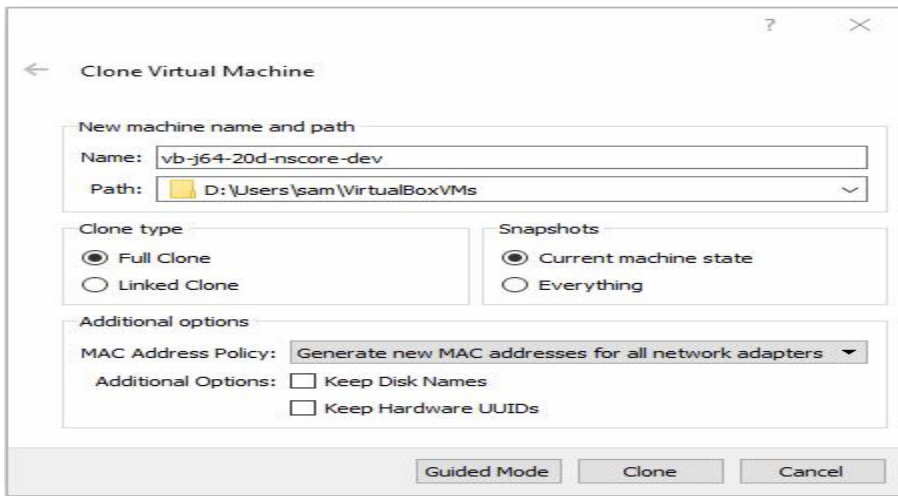


Observe:

- Previous state returns.

Step 6

Clone VM.

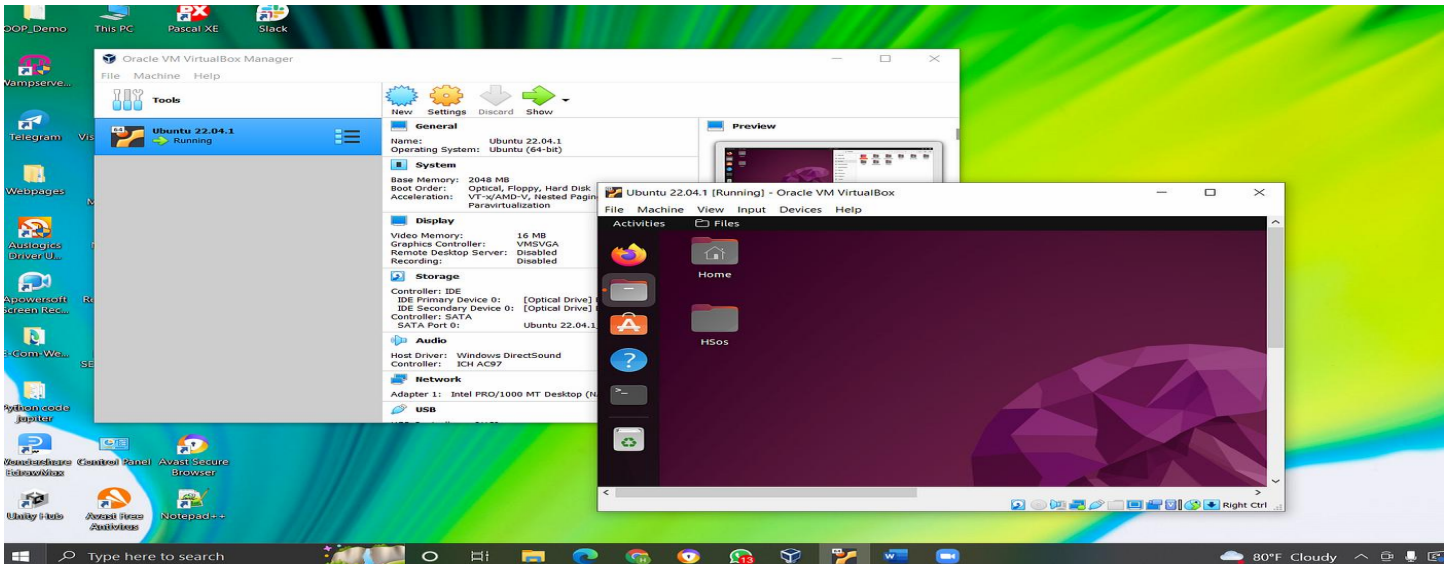


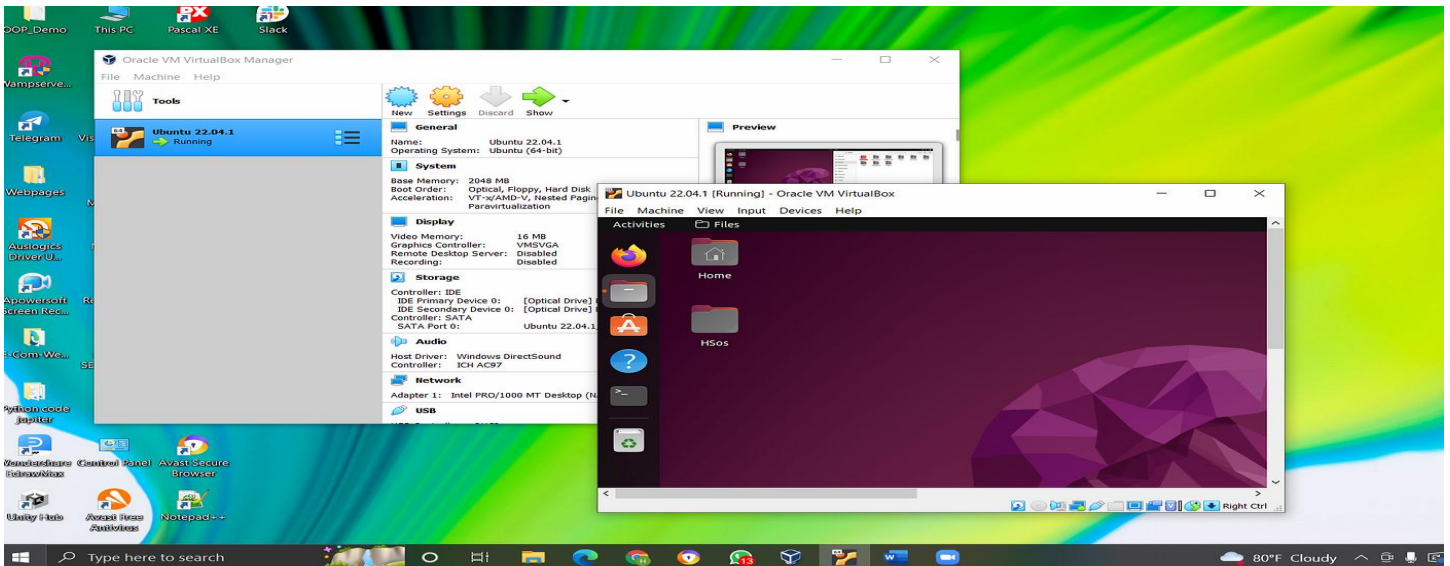
Choose:

- Full Clone

Step 7

Start Cloned VM.





Screenshots to Upload

- Snapshot Creation
- Snapshot List
- Changes Made
- Snapshot Restore
- Clone Wizard
- Cloned VM Running

Result

The Virtual Machine Snapshot and Clone features were successfully demonstrated.